



United International University
Department of Computer Science and Engineering
CSE 1112: Structured Programming Language Laboratory
Assignment, Total Marks: 15

Instructions: Write down C programs for the following problems in Code Blocks (or any C compiler you prefer), copy the code in a document and write line by line explanation on the code you have done. Convert the document into a PDF and submit it on elms.

Any forms of cheating or plagiarism is strictly prohibited, and if found will be penalized.

There are two questions. You have to answer both of them.

Q1

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You are developing a **smart agriculture monitoring system** that records daily soil temperatures (in °C) for 7 consecutive days across 5 different farms in Bangladesh to help farmers analyze environmental conditions for better crop management. Your task is to write a C program that processes this data by storing the temperature readings in a 2D array, where each row represents a farm and each column represents a day. While taking input, the program must ensure that every temperature value lies within the valid range of 5°C to 60°C; if an invalid value is entered, it should display a warning and prompt the user to re-enter the value. After collecting the data, the program should calculate the average temperature for each farm and classify the farm's condition as "High Heat Stress" if the average is 30°C or above, "Optimal" if it falls between 20°C and 29.9°C, and "Low Temperature Stress" if it is below 20°C. Additionally, the program should count how many days each farm experienced extreme conditions, specifically days with temperatures above 40°C (Heat Stress Days) and below 10°C (Cold Stress Days). The program should then display a farm-wise summary including the average temperature, condition classification, and counts of stress days. Furthermore, it should identify and report the farm that recorded the highest single-day temperature and the farm with the lowest single-day temperature. Finally, the program should compute the average temperature for each of the 7 days across all farms and determine which day had the highest and lowest average temperatures, reporting them as the hottest and coldest days respectively.

The sample input/ output is given below:

Input	Output
32 34 33 35 36 38 37 28 29 30 31 32 33 34 18 19 20 21 22 23 24 42 41 40 39 38 37 36 12 14 13 15 16 17 18	Farm-wise Summary Farm 1 Average Temperature: 35.00°C Condition: High Heat Stress Heat Stress Days (>40°C): 0 Cold Stress Days (<10°C): 0 Farm 2 Average Temperature: 31.00°C

	Condition: High Heat Stress Heat Stress Days (>40°C): 0 Cold Stress Days (<10°C): 0 Farm 3 Average Temperature: 21.00°C Condition: Optimal Heat Stress Days (>40°C): 0 Cold Stress Days (<10°C): 0 Farm 4 Average Temperature: 39.00°C Condition: High Heat Stress Heat Stress Days (>40°C): 2 Cold Stress Days (<10°C): 0 Farm 5 Average Temperature: 15.00°C Condition: Low Temperature Stress Heat Stress Days (>40°C): 0 Cold Stress Days (<10°C): 0 Extreme Temperature Records Highest Single-Day Temperature: 42°C (Farm 4) Lowest Single-Day Temperature: 12°C (Farm 5) Daily Average Temperatures (Across All Farms) Day 1: 26.4°C Day 2: 27.4°C Day 3: 27.2°C Day 4: 28.2°C Day 5: 28.8°C Day 6: 29.6°C Day 7: 29.8°C Overall Day Analysis Hottest Day (Highest Average): Day 7 (29.8°C) Coldest Day (Lowest Average): Day 1 (26.4°C)	
Q2	Write about the most interesting problem you have solved during the SPL course. Explain the coding logic on how you were able to solve it.	5